Minkyu Jeon

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Research Summary

Ph.D. candidate in Computer Science at Princeton University, advised by Prof. Ellen D. Zhong. I build AI systems for scientific discovery: agentic systems that coordinate validated scientific tools for de novo protein design, and generative and representation learning methods for heterogeneous cryo-EM reconstruction. My work connects model development, high-performance scientific workflows, and rigorous in silico validation so that generative models can be used as reproducible discovery systems rather than isolated samplers.

Interests: scientific agentic systems; AI for biology; de novo protein design; protein generative models; diffusion and flow matching in 3D structure spaces; cryo-EM heterogeneity; representation learning; 3D vision.

Research Themes

Scientific agents	Closed-loop systems that synthesize structured scientific evidence, allocate GPU compute across tools, and keep decisions auditable.
Protein design	De novo binder generation, interface-confidence scoring, structural uniqueness, sequence novelty, and wet-lab candidate-panel selection.
Cryo-EM	Generative and amortized methods for continuous conformational heterogeneity, with benchmarks that stress-test reconstruction quality and robustness.

Education

Princeton University *2023–present*
Ph.D. Candidate, Computer Science — Princeton, NJ

Korea University *2020–2023*
M.S., Computer Science and Engineering — Seoul, Korea

Konkuk University *2014–2020*
B.S., Applied Statistics and Computer Engineering, double major — Seoul, Korea

Selected Research Projects

T-ReX: Agentic Inference-Time Scaling for De Novo Protein Design *2026–present*
Princeton University

- Developing a closed-loop binder-design system that treats de novo generation as target-adaptive compute allocation across published design tools, redesign actions, and canonical scoring.
- Designed an LLM Planner/Supervisor interface that reads structured EvidenceSummary records, proposes Rescue–Explore–eXploit actions, and leaves final action validation and GPU scheduling to deterministic software.
- Built evaluation workflows for hard design targets, including strict success criteria, Foldseek structural uniqueness, MMseqs2 sequence novelty, AF2 ipTM diagnostics, and post hoc candidate-panel ranking.
- Emphasis: produce more structure-unique, in silico qualified binders per GPU-hour while preserving scientific evidence trails for analysis and wet-lab prioritization.

Heterogeneous Cryo-EM Reconstruction *2023–present*
Princeton University

- Developing amortized, self-distilled, and transformer-based methods for reconstructing continuous conformational heterogeneity from cryo-EM particle images.
- Led CryoBench, a benchmark suite for diverse and challenging cryo-EM heterogeneity problems, accepted as a NeurIPS 2024 Spotlight.
- Co-led CryoHype, a transformer-hypernetwork approach for reconstructing large conformational ensembles, accepted to CVPR 2026.

Publications

* denotes equal contribution.

CVPR 2026 — **CryoHype: Reconstructing a thousand cryo-EM structures with transformer-based hypernetworks**
Jeffrey Gu*, **Minkyu Jeon***, Ambri Ma, Serena Yeung-Levy, Ellen D. Zhong
paper

arXiv 2026	**ConforNets: Latents-Based Conformational Control in OpenFold3** Minji Lee, Colin Kalicki, **Minkyu Jeon**, Aymen Qabel, Alisia Fadini, Mohammed AlQuraishi paper
NeurIPS MLSB 2025	**Separating signal from noise: a self-distillation approach for amortized heterogeneous cryo-EM reconstruction** **Minkyu Jeon***, Jeffrey Gu*, Ambri Ma, Serena Yeung-Levy, Vincent Sitzmann, Ellen D. Zhong **Oral presentation**. paper
NeurIPS 2024	**CryoBench: Diverse and challenging datasets for the heterogeneity problem in cryo-EM** **Minkyu Jeon**, Rishwanth Raghu, Miro Astore, Geoffrey Woollard, Ryan Feathers, Alkin Kaz, Sonya M. Hanson, Pilar Cossio, Ellen D. Zhong **Spotlight**. paper / project
NeurIPS 2022	**SageMix: Saliency-Guided Mixup for Point Clouds** Sanghyeok Lee*, **Minkyu Jeon***, Injae Kim, Yunyang Xiong, Hyunwoo J. Kim paper / code
ECCV 2022	**k-SALSA: k-anonymous synthetic averaging of retinal images via local style alignment** **Minkyu Jeon**, Hyeonjin Park, Hyunwoo J. Kim, Michael Morley, Hyunghoon Cho paper / code
JMIR 2024	**A multivariable prediction model for mild cognitive impairment and dementia: algorithm development and validation** Sarah Soyeon Oh, Bada Kang, Dahye Hong, Jennifer Ivy Kim, Hyewon Jeong, Jinyeop Song, **Minkyu Jeon**
Information Sciences 2023	**Randomly shuffled convolution for self-supervised representation learning** Youngjin Oh*, **Minkyu Jeon***, Dohwan Ko, Hyunwoo J. Kim paper / code
IEEE Access 2021	**Learning to Balance Local Losses via Meta-Learning** Seungdong Yoa, **Minkyu Jeon**, Youngjin Oh, Hyunwoo J. Kim paper
In submission	**CHIMERA: Adaptive Cache Injection and Semantic Anchor Prompting for Zero-shot Image Morphing with Morphing-oriented Metrics** Dahyeon Kye, Jaehun Sung, **Minkyu Jeon**, Jihyong Oh paper / project
arXiv 2025	**R3eVision: A Survey on Robust Rendering, Restoration, and Enhancement for 3D Low-Level Vision** Weeyoung Kwon, Jeahun Sung, **Minkyu Jeon**, Chanho Eom, Jihyong Oh paper

Research Experience

Prescient Design, Genentech — *Nov 2025–Mar 2026*
Contractor — New York, NY

- Worked on protein representation learning and generative modeling for protein design, with emphasis on modern 3D generative models including diffusion and flow matching.
- Studied limitations of voxel- and structure-space generative models and developed evaluation workflows for diagnosing and improving generation performance.

Prescient Design, Genentech — *Jun 2025–Aug 2025*
Research Intern — New York, NY

- Led a project on understanding and improving modern generative models in 3D spaces, with applications to protein generation and design.
- Connected model behavior to practical design diagnostics, including structure quality, interface plausibility, and robustness across target settings.

AI4ALL — *Jul 2024–Aug 2024*
Instructor — Princeton, NJ

- Instructed high school students in AI and mentored team projects on medical image analysis using deep learning.

Broad Institute of MIT and Harvard *Jan 2023–Aug 2023*
Associate Computational Biologist; advisor: Dr. Hyunghoon Cho Cambridge, MA

- Led a project on synthesizing privacy-preserving retinal image datasets with diffusion-based generative models.

Broad Institute of MIT and Harvard *Sep 2021–Apr 2022*
Visiting Graduate Student; advisor: Dr. Hyunghoon Cho Cambridge, MA

- Led GAN and GAN-inversion based retinal image synthesis research for privacy-preserving data generation, resulting in an ECCV 2022 publication.

Korea University *Jan 2020–Aug 2020*
Research Intern; advisor: Prof. Hyunwoo J. Kim Seoul, Korea

- Conducted meta-learning research for layer-wise optimization and co-authored an IEEE Access publication.
- Designed algorithms, implemented experiments, and contributed to related work, method, and experiment sections.

MakinaRocks *Mar 2019–Aug 2019*
ML Research Engineer Intern Seoul, Korea

- Conducted anomaly detection research using variational autoencoders and latent-space regularization.

Korea Institute of Science and Technology *Jul 2018–Dec 2018*
ML Research Intern; advisor: Dr. Yong Moo Kwon Seoul, Korea

- Worked on an active chatbot project using rule-based NLP and deep learning methods for situation-aware conversation.

Awards and Service

2024–present	**Asan Foundation Biomedical Science Scholarship**. Three-year doctoral scholarship.
2023–present	**Reviewer**. CVPR (2026), ECCV (2026), ICLR (2025, 2026), NeurIPS (2025, 2026), NeurIPS Workshop (2023, 2024), Information Sciences (2024).
Aug 2024	**Invited talk**. Flatiron Institute, Simons Foundation.
Feb 2024	**Invited talk**. CMU MetaMobility Lab.
Sep 2019	**Dean's List**. Department of Applied Statistics, Konkuk University.
Sep 2019	**Konkuk University Scholarship**. 40% tuition scholarship.
Jul 2016	**Certificate of Commendation**. Republic of Korea Marine Corps.

Teaching

Fall 2024	Teaching Assistant, COS302: Mathematics for Numerical Computing and Machine Learning, Princeton University.
Spring 2021	Teaching Assistant, COSE361: Artificial Intelligence, Korea University.
Fall 2020	Teaching Assistant, AAA712: AI Security, Korea University.
Fall 2020	Tutorial Teaching Fellow, Deep Learning, Korea University.

Skills

Programming	Python, PyTorch, TensorFlow, Keras, R, C, Git.
Systems	Linux, Slurm/HPC workflows, reproducible experiment pipelines, scientific data processing, macOS, Windows.
Scientific ML	LLM-guided scientific workflows, generative modeling, diffusion/flow matching, protein design, cryo-EM reconstruction, representation learning, 3D vision.
Bio/structure tools	Protein design evaluation, AlphaFold-style confidence diagnostics, Foldseek, MMseqs2, structural clustering, candidate-panel analysis.

Additional Experience

I-Corps Program *May 2021–Aug 2021*
Received $35,000 in government support for customer discovery and technology-based startup exploration.

Republic of Korea Marine Corps *Mar 2015–Dec 2016*
Military service, Yeonpyeong Island, South Korea; received commendation for communication and performance during early warning situations.